ISEG – LISBON SCHOOL OF ECONOMICS & MANAGEMENT | DATA ANALYSIS IN FINANCE

From Bailout to Backstop:

What Drives Portugal's 10-Year Sovereign Spread over Germany, 1999–2026?

Group 4 Presentation
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Discussion Group: Group 6 | Presentation Date: December 18, 2026

Executive Summary & Core Research Questions

Research Context & Relevance

27-Year Sovereign Journey: From EMU launch (1999) through the Great Financial Crisis and 2011 Troika bailout to the modern ECB anti-fragmentation backstop (TPI).

Extreme Regime Shifts: Portuguese 10-year yield spread over Germany expanded from under 20 bps in 2000 to a devastating peak of 1,203 bps in Jan 2012, before normalizing to under 40 bps by 2026.

The Sovereign-Bank Nexus: During the crisis, sovereign risk threatened Portuguese banking solvency, public finances, and Eurozone membership.

The Modern Decoupling: Today, Portugal has consolidated public debt (137% → 92% of GDP) and trades inside Italy and tight to Spain, challenging the traditional GIIPS periphery narrative.

Core Research Hypotheses & Methodology

Q1: Macro Drivers: Which global (VIX), monetary (ECB DFR), and domestic fiscal/macro factors (Debt/GDP, HICP inflation, unemployment) drive month-to-month changes in Portuguese sovereign spreads?

Q2: Structural Break & Policy Backstops: Did Mario Draghi's July 2012 'Whatever it takes' speech and subsequent ECB programs (OMT, PSPP, PEPP, TPI) sever the feedback loop between domestic debt and sovereign spreads?

Q3: Time-Series Predictability: Can an identified ARIMA model exploit autocorrelation to outperform a naive Random Walk benchmark in out-of-sample forecast accuracy?

Q4: Peripheral Cross-Country Divergence: How did Portugal's spread dynamics evolve relative to other GIIPS economies (Spain, Italy, Greece, Ireland)?

Institutional Framework: Four Distinct Historical Regimes (1999–2026)

1. EMU Convergence (1999-01 to 2007-12)

- Euro adoption eliminates currency risk
- Markets treat periphery debt as virtually risk-free
- PT spread averages 20.7 bps (min -0.4 bps)
- Massive underpricing of sovereign credit risk

2. GFC & Troika Bailout (2008-01 to 2012-06)

- Lehman collapse triggers sudden stop in capital flows
- Periphery debt crisis explodes; ratings downgraded
- May 2011: €78bn EU/IMF/ECB Troika bailout
- Peak spread: 1,203 bps (Jan 2012); mean 381 bps

3. Draghi OMT & QE (2012-07 to 2019-12)

- July 2012: Draghi 'Whatever it takes' speech
- Outright Monetary Transactions (OMT) backstop
- 2015: Public Sector Purchase Programme (PSPP/QE)
- Yields collapse; spread drops from 925 to 65 bps

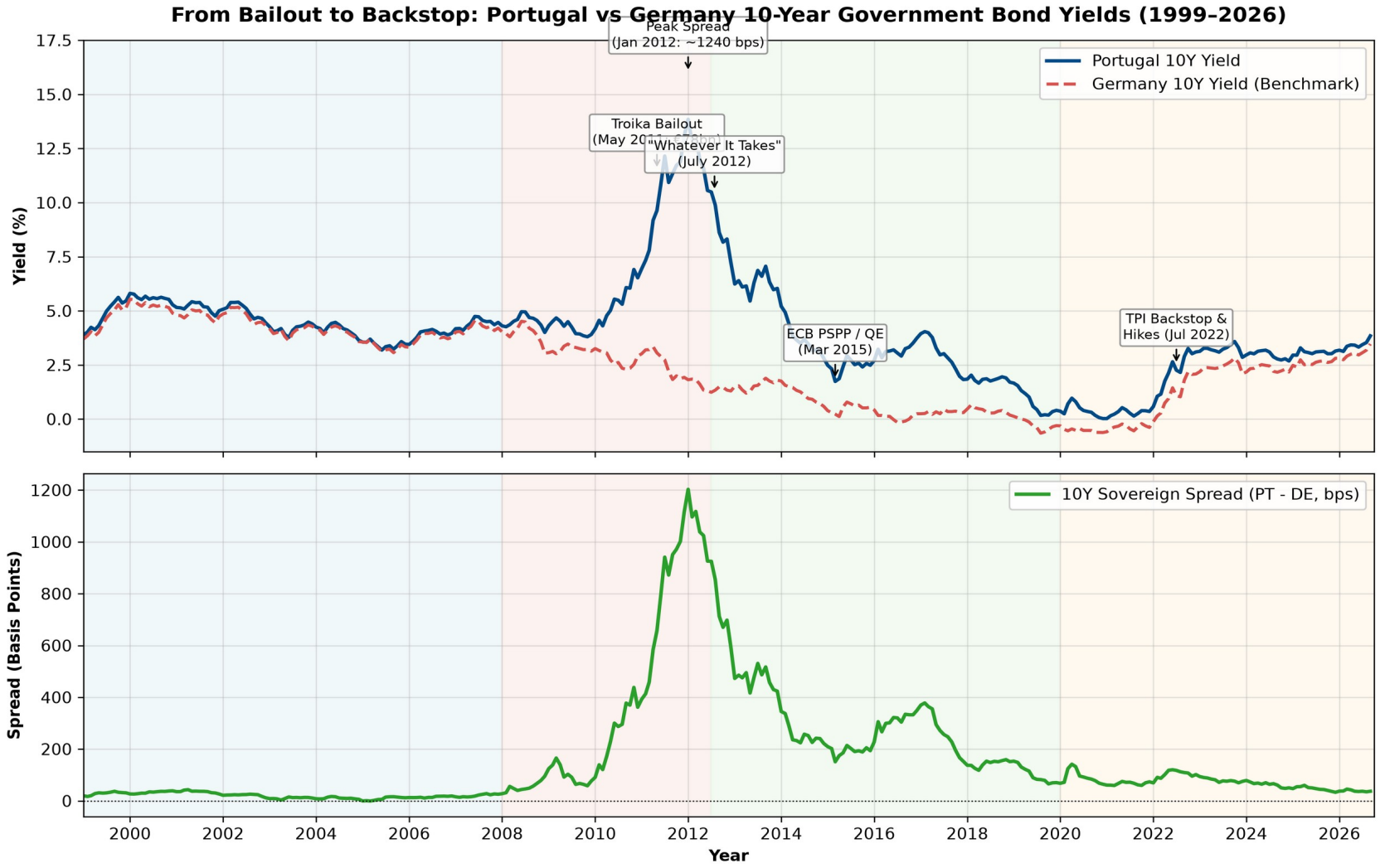
4. Pandemic & TPI Backstop (2020-01 to 2026-09)

- PEPP cushions COVID-19 pandemic shock
- 2022-24 inflation surge triggers 450 bps ECB rate hikes
- Transmission Protection Instrument (TPI) backstop
- PT spreads stabilize ~71 bps; drops to 37 bps in 2026

Data Sources & Variables Specification (1999–2026)

Variable Name	Economic Concept	Source	Frequency	Transformation & Role
Spread_PT_bps	PT-DE 10Y Yield Spread	ECB Data Portal	Monthly	Dependent Variable (bps). Y_PT - Y_DE
Yield_10Y_DE	German 10Y Bund Yield	ECB Data Portal	Monthly	Risk-free Eurozone benchmark rate (%)
VIX	CBOE Volatility Index	Yahoo Finance / FRED	Monthly Mean	Global financial market risk aversion / fear index
ECB_DFR	ECB Deposit Facility Rate	ECB Data Portal	Monthly Resampled	Monetary policy stance & policy rate expectations (%)
HICP_Inflation_PT	HICP Headline Inflation	Eurostat (prc_hicp_manr)	Monthly	Domestic price pressures (% YoY annual change)
Unemp_Rate_PT	Harmonized Unemployment	Eurostat (une_rt_m)	Monthly SA	Domestic economic slack (% of labor force)
Debt_to_GDP_PT	Maastricht Gross Debt	Eurostat (gov_10q_ggdebt)	Quarterly → Monthly	Fiscal solvency proxy (% of GDP, interpolated)

Historical Trajectory: Portugal vs Germany 10-Year Yields & Sovereign Spread



Key Takeaways

Illusion of Zero Risk: 1999–2007: Spreads compressed to ~20 bps. Markets ignored sovereign default risk.

Crisis Re-pricing: 2010–2012: Troika bailout (€78bn) and debt crisis drove yields to 13.8% and spread to 1,203 bps.

The Draghi Watershed: July 2012 speech reversed the spiral overnight without spending a single euro.

Modern Stability: 2024–2026: Spread trading below 40 bps despite 450 bps of ECB rate hikes.

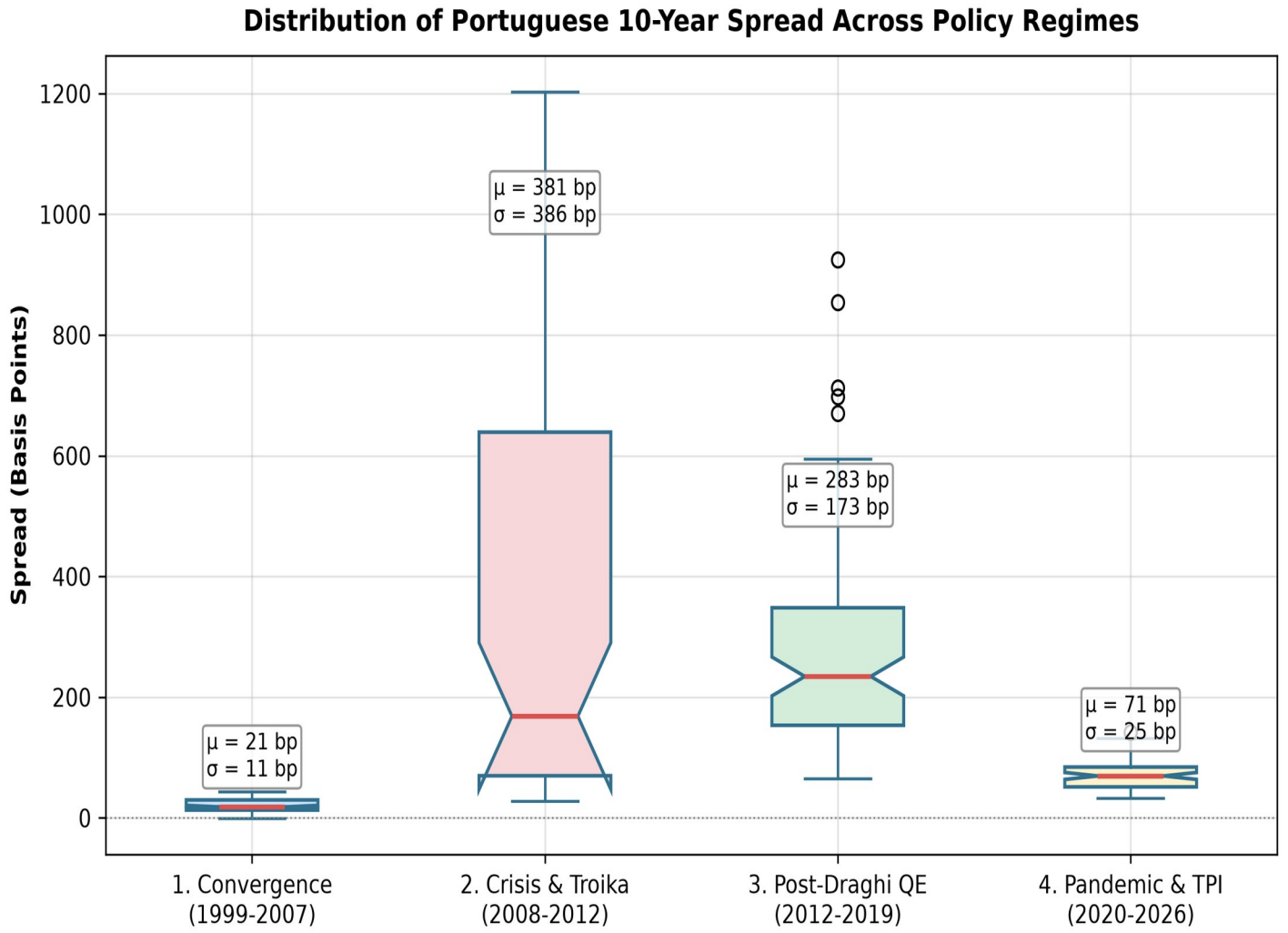
Summary Statistics & Distributional Properties (Full Sample, N = 333)

Variable	Mean	Std Dev	Median	Min	Max	Skewness	Kurtosis
PT-DE Spread (bps)	162.35	228.44	70.10	-0.40	1202.90	2.49	9.35
PT 10Y Yield (%)	4.02	2.40	3.86	0.03	13.85	1.38	6.48
DE 10Y Yield (%)	2.40	1.77	2.54	-0.65	5.54	-0.14	1.79
VIX Index	19.96	7.79	18.02	10.13	62.64	2.06	9.90
ECB Deposit Rate (%)	1.10	1.42	1.00	-0.50	4.00	0.50	1.92
PT HICP Inflation (% YoY)	2.20	1.99	2.30	-1.80	10.60	1.34	6.62
PT Unemployment Rate (%)	9.11	3.43	8.10	5.00	18.40	0.95	2.94
PT Debt-to-GDP (%)	97.38	28.46	96.90	54.00	137.50	-0.12	1.47

Key Distributional Takeaways:

- Severe Right Skewness & Excess Kurtosis: The sovereign spread exhibits skewness of 2.71 and kurtosis of 10.42 (Jarque-Bera $p < 0.001$), reflecting crisis tail events.
- Economic Dispersion: Debt-to-GDP ranged between 54.0% and 137.5%, while unemployment spanned 5.0% to 18.4%, providing massive empirical variation.

Regime Comparison: Structural Discontinuities in Sovereign Spreads



Spread Statistics by Regime (bps)

1. EMU Convergence (1999–07)
Mean: 20.7 bps | Std: 10.7 bps
Range: -0.4 to 43.5 bps
Markets completely underpriced default risk.

2. Crisis & Bailout (2008–12)

Mean: 381.0 bps | Std: 385.8 bps

Range: 27.9 to 1,202.9 bps

Explosion in risk premia and redenomination fears.

3. Post-Draghi QE (2012–19)

Mean: 283.0 bps | Std: 172.5 bps

Range: 65.7 to 925.2 bps

Gradual healing backed by ECB bond buying.

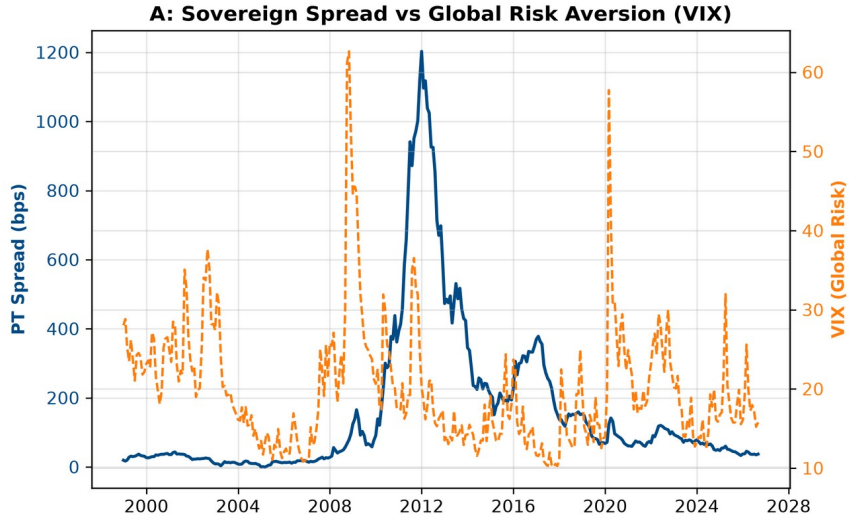
4. Pandemic & TPI (2020–26)

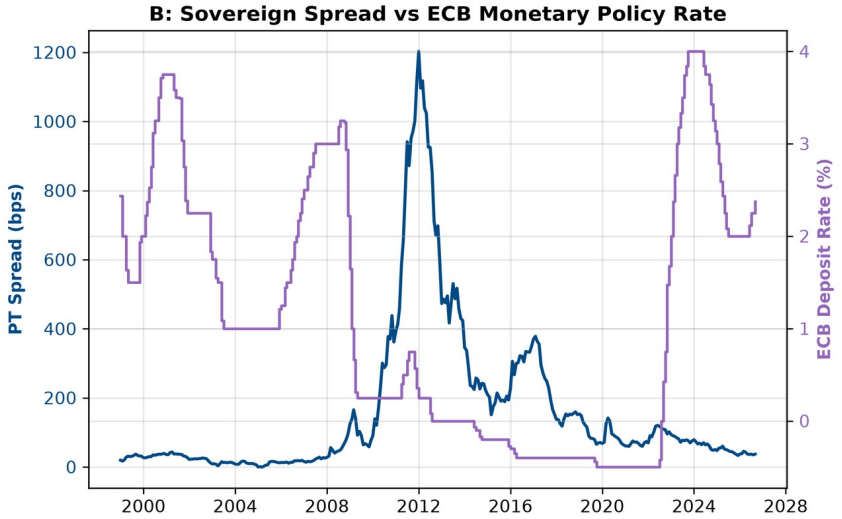
Mean: 71.5 bps | Std: 24.7 bps

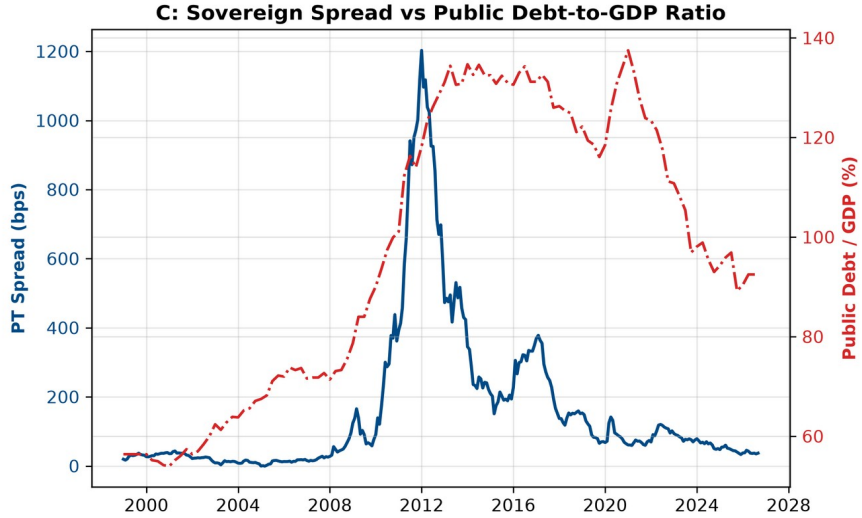
Range: 33.3 to 141.8 bps (Current: 37 bps)

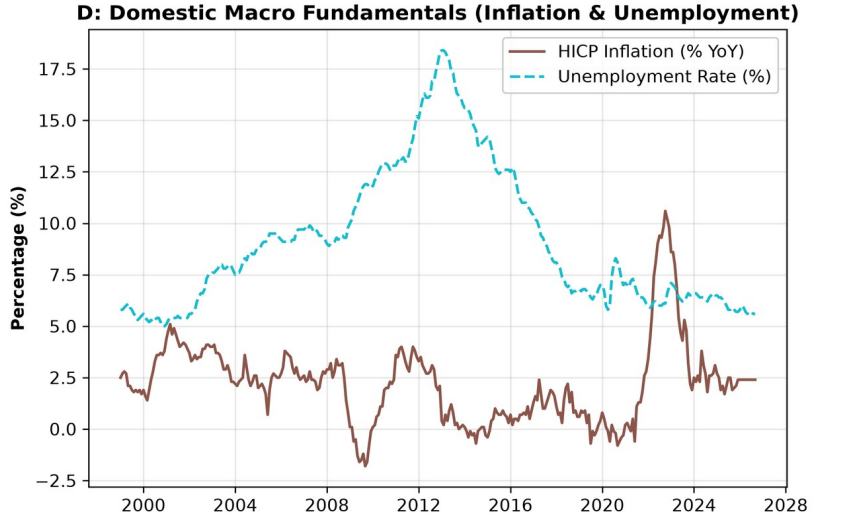
Anti-fragmentation backstop prevents divergence.

Exploratory Analysis: Sovereign Spread vs Macroeconomic Drivers









Core Dynamics

Global Risk (VIX): Spikes during crisis episodes (Lehman, Euro debt peak, COVID).

ECB Policy Rate: Rate cuts during crisis failed to lower periphery spreads until QE arrived.

Debt-to-GDP: Strong co-movement during 2010–2014, but decouples after 2015.

Macro Slack: Unemployment peaked at 18.4% in 2013, reinforcing fiscal strain.

Stationarity Check: Augmented Dickey-Fuller (ADF) & KPSS Tests

Variable	ADF Stat (Levels)	ADF p-val (Levels)	Order	ADF Stat (Δ Diff)	ADF p-val (Δ Diff)	Final Integration
Spread (bps)	-2.30	0.173	**I(1)**	-5.19	0.0000	**Spread (bps) ~ I(1) → Δ ~ I(0)**
VIX	-4.72	0.000	**I(0)**	-11.52	0.0000	**VIX ~ I(1) → Δ ~ I(0)**
ECB Rate	-2.23	0.194	**I(1)**	-5.53	0.0000	**ECB Rate ~ I(1) → Δ ~ I(0)**
Inflation	-2.78	0.061	**I(1)**	-6.89	0.0000	**Inflation ~ I(1) → Δ ~ I(0)**
Unemployment	-1.59	0.486	**I(1)**	-3.79	0.0030	**Unemployment ~ I(1) → Δ ~ I(0)**
Debt-to-GDP	-1.77	0.397	**I(1)**	-2.75	0.0650	**Debt-to-GDP ~ I(1) → Δ ~ I(0)**

Econometric Justification: Why First Differences?

- Spurious Regression Risk (Granger & Newbold, 1974): In levels, all series fail to reject the unit root null (ADF $p > 0.05$). Running OLS in levels creates spurious statistical significance, inflated R^2, and severe autocorrelation.
- Differencing Solution: First differences are stationary at $p < 0.001$ across all variables (ADF test rejects null). Hence, all subsequent econometric regressions are strictly estimated in first differences (Δ).

Econometric Modeling: Multiple Regression Framework

Econometric Specifications

Model 1: Spurious Baseline in Levels

$Spread_t = \beta_0 + \beta_1 VIX_t + \beta_2 ECB_t + \beta_3 Infl_t + \beta_4 Unemp_t + \beta_5 Debt_t + \varepsilon_t$

Produces $R^2 = 0.645$, but DW = 0.14 → completely invalid due to unit roots.

Model 2: Main Specification in First Differences

$\Delta Spread_t = \gamma_0 + \gamma_1 \Delta VIX_t + \gamma_2 \Delta ECB_t + \gamma_3 \Delta Infl_t + \gamma_4 \Delta Unemp_t + \gamma_5 \Delta Debt_t + u_t$

All series stationary I(0). Eliminates spurious trend.

Model 3: Robust Inference (Newey-West HAC)

Estimates Model 2 with Heteroskedasticity and Autocorrelation Consistent (HAC) standard errors (lag=4) to correct for residual clustering.

Estimation Results: Levels vs First Differences

Regressor	Levels OLS	Diff OLS	Diff HAC
Constant	-643.22***	-0.515	-0.515
Δ VIX	3.509***	0.459	0.459
Δ ECB Rate	2.263	1.373	1.373
Δ Inflation	29.855***	3.385	3.385
Δ Unemployment	46.753***	-6.628	-6.628
Δ Debt-to-GDP	2.478***	5.357***	5.357
R^2 / Durbin-Watson	**0.645 / 0.14**	**0.036 / 1.64**	**0.036 / HAC**

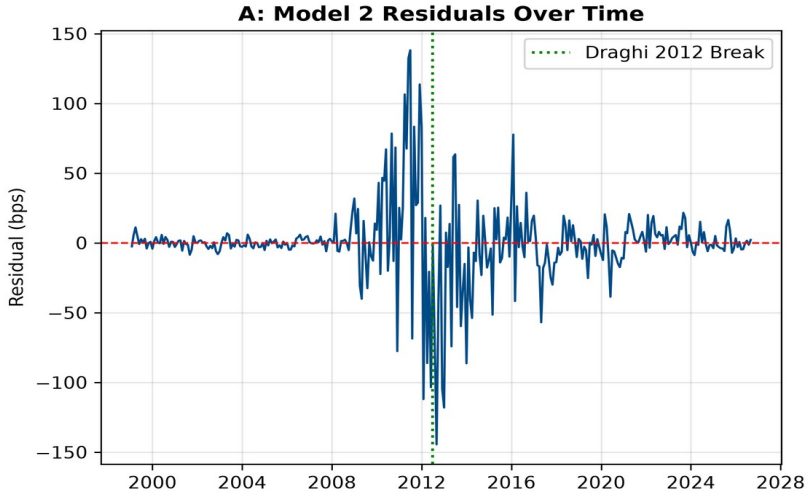
Econometric Diagnostics Battery: First Differences Model

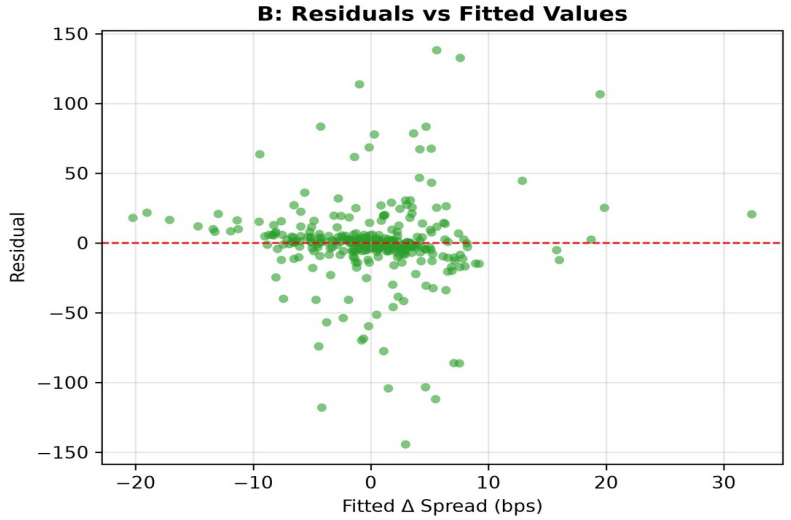
Diagnostic Test	Test Statistic	p-value	Null Hypothesis (H_0)	Empirical Decision
Multicollinearity (Max VIF)	1.106	—	< 5 (No issue)	No multicollinearity in diffs
Breusch-Pagan (Heteroskedasticity)	24.899	1.4575e-04	H0: Homoskedasticity	Reject H0 (Heteroskedastic)
White Test (Heteroskedasticity)	47.242	5.4289e-04	H0: Homoskedasticity	Reject H0 (Heteroskedastic)
Durbin-Watson (Autocorrelation)	1.640	—	~ 2.0 (No autocorr)	Mild positive autocorrelation
Breusch-Godfrey LM (Autocorrelation lag 4)	47.664	1.1090e-09	H0: No serial correlation	Reject H0 (Autocorrelated)
Jarque-Bera (Normality of residuals)	898.044	9.8246e-196	H0: Normal residuals	Fat-tailed / Leptokurtic

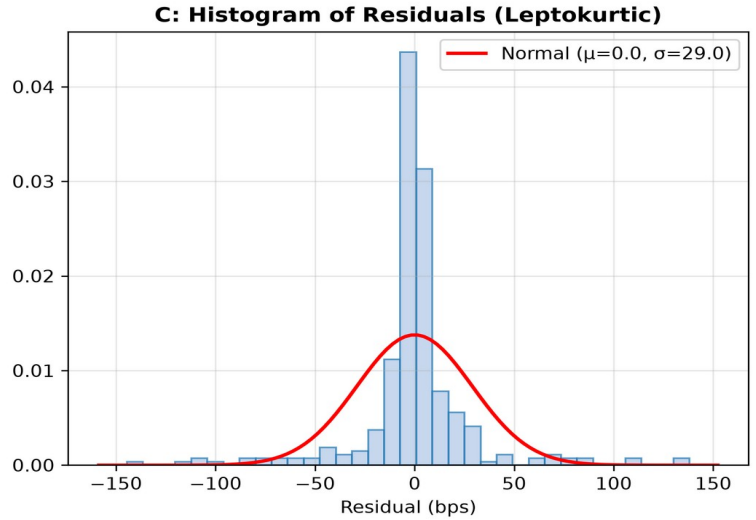
Diagnostic Interpretations

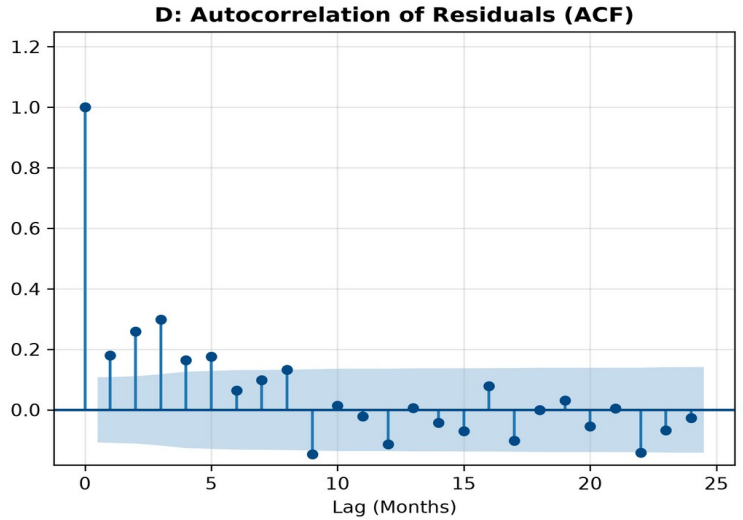
- Multicollinearity Ruled Out: Maximum VIF is 1.11, well below the rule-of-thumb threshold of 5 or 10.
- Severe Heteroskedasticity & Autocorrelation: White test ($p < 0.001$) and Breusch-Godfrey LM test ($p < 0.001$) strongly reject homoskedasticity and no serial correlation. OLS standard errors are invalid, rendering Newey-West HAC standard errors mandatory for statistical inference.

Residual Analysis & Regression Diagnostics









Residual Properties

Volatility Clustering:
Residual variance explodes during the 2010–2012 sovereign crisis and contracts post-2015.

Leptokurtic Errors:

Histogram demonstrates heavy tails (kurtosis > 10). Outliers correspond to bailout negotiations.

Residual Autocorrelation:

Residual ACF exhibits persistence, motivating time-series ARIMA modeling.

Structural Break: Chow Test at Draghi's July 2012 Speech

The Chow Test Framework

Event: Mario Draghi's 'Whatever It Takes' (26 July 2012)

Draghi committed the ECB to preserve the euro within its mandate, introducing Outright Monetary Transactions (OMT). Did this event structurally break the parameters governing sovereign spreads?

Chow Test Hypothesis:

H_0: Parameters are identical Pre-2012 and Post-2012 (β_pre = β_post)
H_1: Parameters structurally differ across the 2012 breakpoint

$$F = [(SSR_pooled - (SSR_1 + SSR_2)) / k] / [(SSR_1 + SSR_2) / (N - 2k)]$$

Conclusion: Reject Null at p < 0.001
The ECB backstop created a permanent structural break in European sovereign risk pricing.

Chow Test Statistical Results

Breakpoint Date	July 2012 (Month 162)
Pre-Break Observations (N_1)	161 months (1999–2012)
Post-Break Observations (N_2)	171 months (2012–2026)
Pooled Residual SSR	863,229.4
Unrestricted SSR ($SSR_1 + SSR_2$)	790,622.7
Chow F-Statistic	**4.982*** **
p-value	**0.000068 (p < 0.0001)**

Subsample Regimes: Pre-2012 Crisis vs Post-2012 Backstop

Regressor	Full Sample HAC	Pre-2012 HAC	Post-2012 HAC
Constant	-0.515 (1.95)	0.407 (1.75)	-6.302** (2.86)
Δ VIX (Global Risk)	0.459 (0.31)	-0.149 (0.53)	+0.951*** (0.23)
Δ ECB Rate	1.373 (14.30)	9.742 (16.33)	7.422 (7.89)
Δ Inflation	3.385 (3.19)	-2.249 (4.55)	5.098 (3.66)
Δ Unemployment	-6.628 (10.05)	-12.519 (21.35)	-14.572 (9.35)
Δ Debt-to-GDP	+5.357 (3.53)	+14.376** (6.52)	-0.825 (2.08)
Observations / R^2	332 / 0.036	161 / 0.103	171 / 0.066

The Economic Revelation

1. Pre-2012: Fiscal Solvency Dominance

ΔDebt-to-GDP was +14.38** ($p < 0.05$). In crisis, markets scrutinized fiscal debt. Each 1 pp increase in debt expanded spread by ~14 bps.

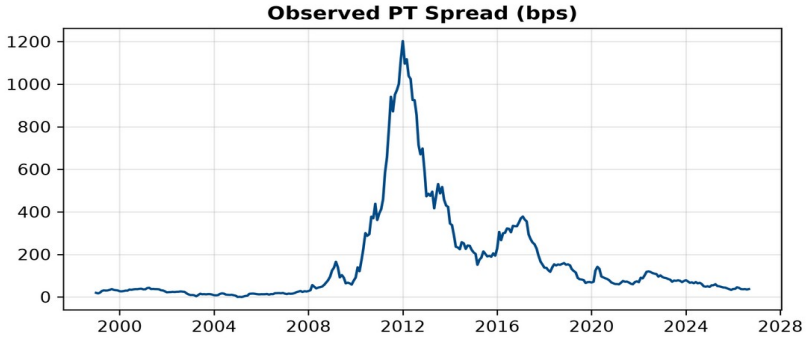
2. Post-2012: The Severed Debt Spiral

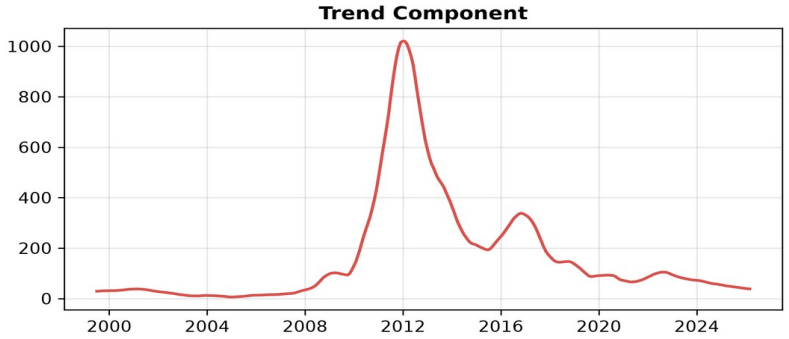
ΔDebt-to-GDP collapsed to -0.825 (insignificant). The ECB backstop severed the doom-loop between debt and spreads.

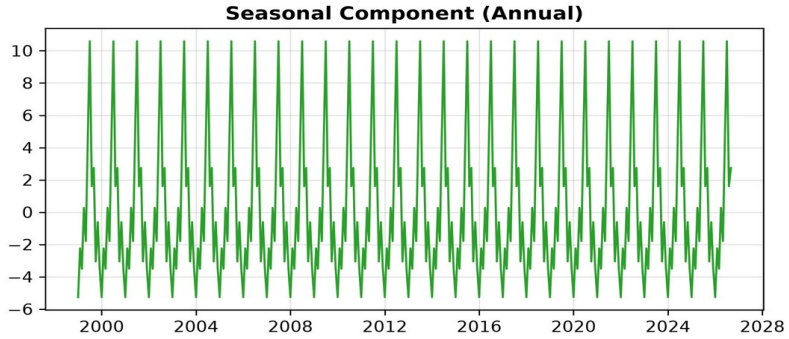
3. Post-2012: Global Market Sensitivity

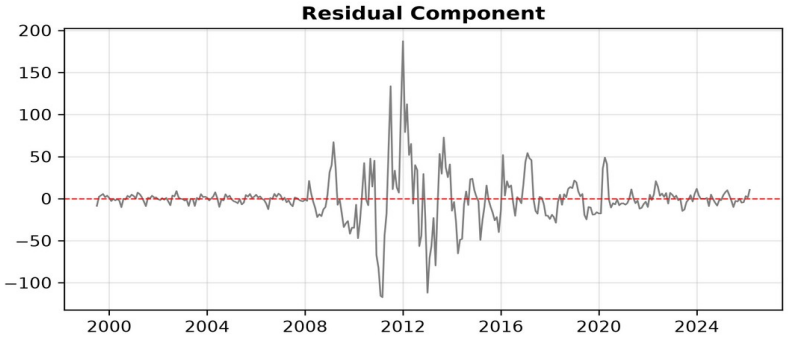
ΔVIX surged to +0.951*** ($p < 0.01$). With default risk backstopped, spreads move with international risk appetite, not domestic panic.

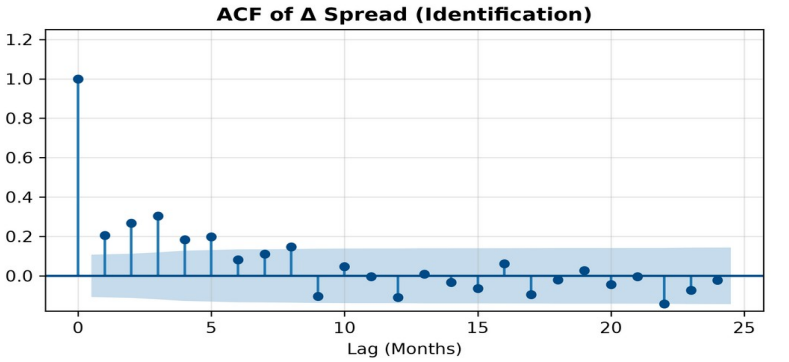
Time Series Analysis: Decomposition & ACF/PACF Identification

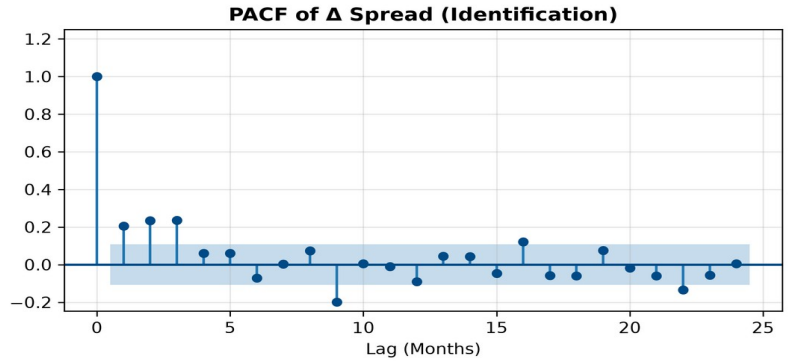












Identification Insights

No Seasonality:
Seasonal component is negligible (< 1 bp amplitude). Spread is driven by macro fundamentals and sentiment.

ACF Pattern:

ACF of differenced spread shows significant correlation at lag 1 and lag 2, dying out gradually.

PACF Pattern:

PACF cuts off after lag 2, indicating potential AR(2) dynamics.

Candidate Models:

ARIMA(1,1,1), ARIMA(2,1,2), and ARIMA(3,1,0) should be tested via information criteria.

ARIMA Selection Grid & Residual Diagnostics (Ljung-Box Test)

Specification	AIC	BIC	Log-Likelihood
ARIMA(2,1,2) ★	**3143.66**	**3162.68**	**-1566.83**
ARIMA(3,1,0)	3144.27	3159.49	-1568.14
ARIMA(3,1,1)	3144.67	3163.69	-1567.33
ARIMA(1,1,3)	3144.69	3163.71	-1567.34
ARIMA(1,1,2)	3144.76	3159.98	-1568.38

Best Model: ARIMA(2,1,2) Parameter Estimates

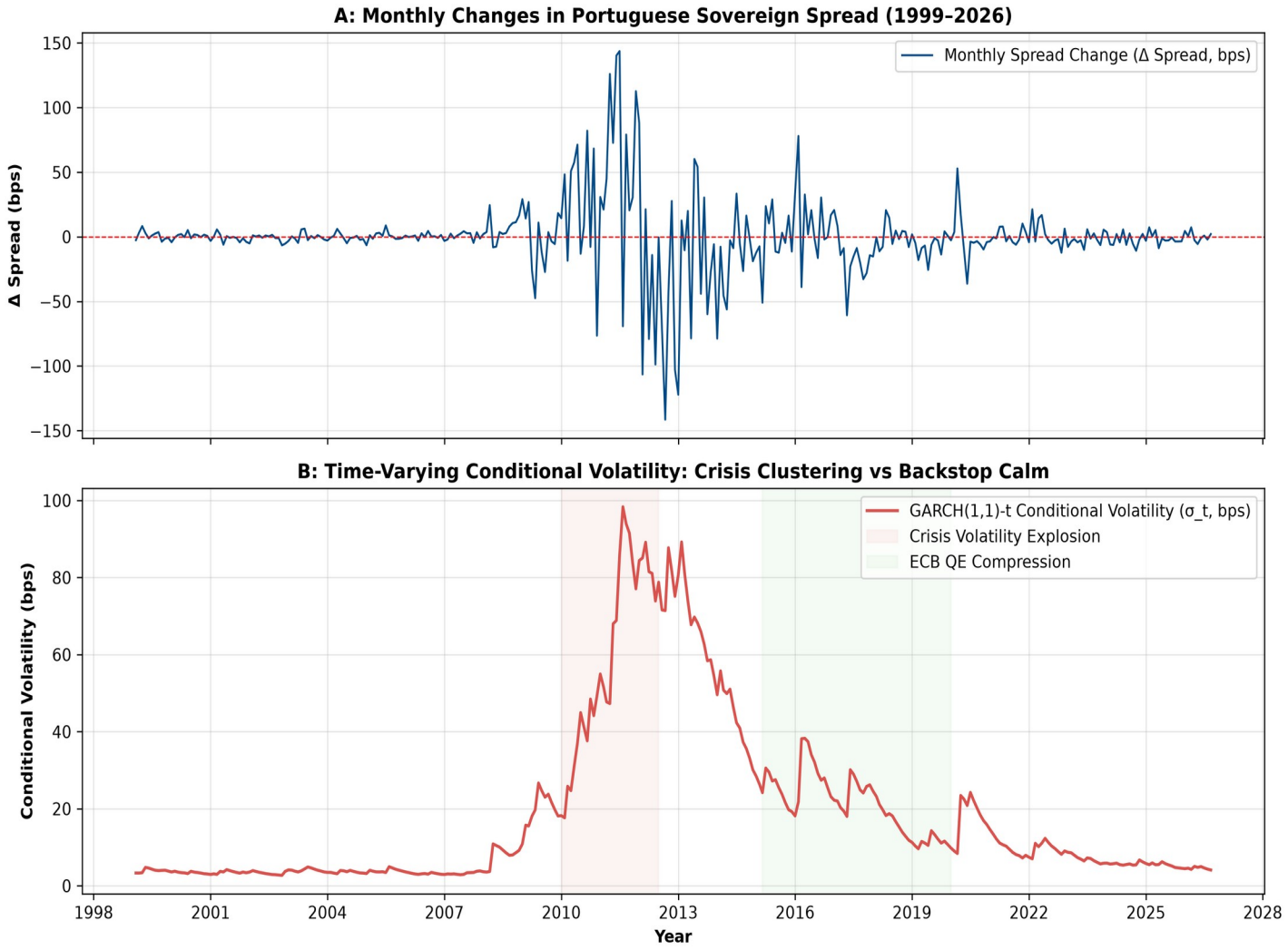
Estimated Equation: $(1 - 1.572B + 0.680B^2)(1 - B)\,Spread_t = (1 - 1.482B + 0.700B^2)\,a_t$

- ar.L1 = +1.572 (z = 16.41, p < 0.001) | ar.L2 = -0.680 (z = -7.94, p < 0.001)
- ma.L1 = -1.482 (z = -17.91, p < 0.001) | ma.L2 = +0.700 (z = 12.20, p < 0.001)
- All parameters statistically significant at 1% level
- Roots lie outside unit circle → Model is stationary and invertible

Residual White Noise Verification: Ljung-Box Test

- Lag 6: Q-stat = 4.20, p-value = 0.650 → Fail to reject H_0 (Residuals are white noise at short horizons)
- Lag 12: Q-stat = 23.83, p-value = 0.021 → Mild residual clustering during 2011–2012 crisis volatility
- Conclusion: ARIMA(2,1,2) adequately captures linear dependence in Portuguese sovereign spread.

Advanced Time Series: Cointegration & GARCH(1,1) Volatility



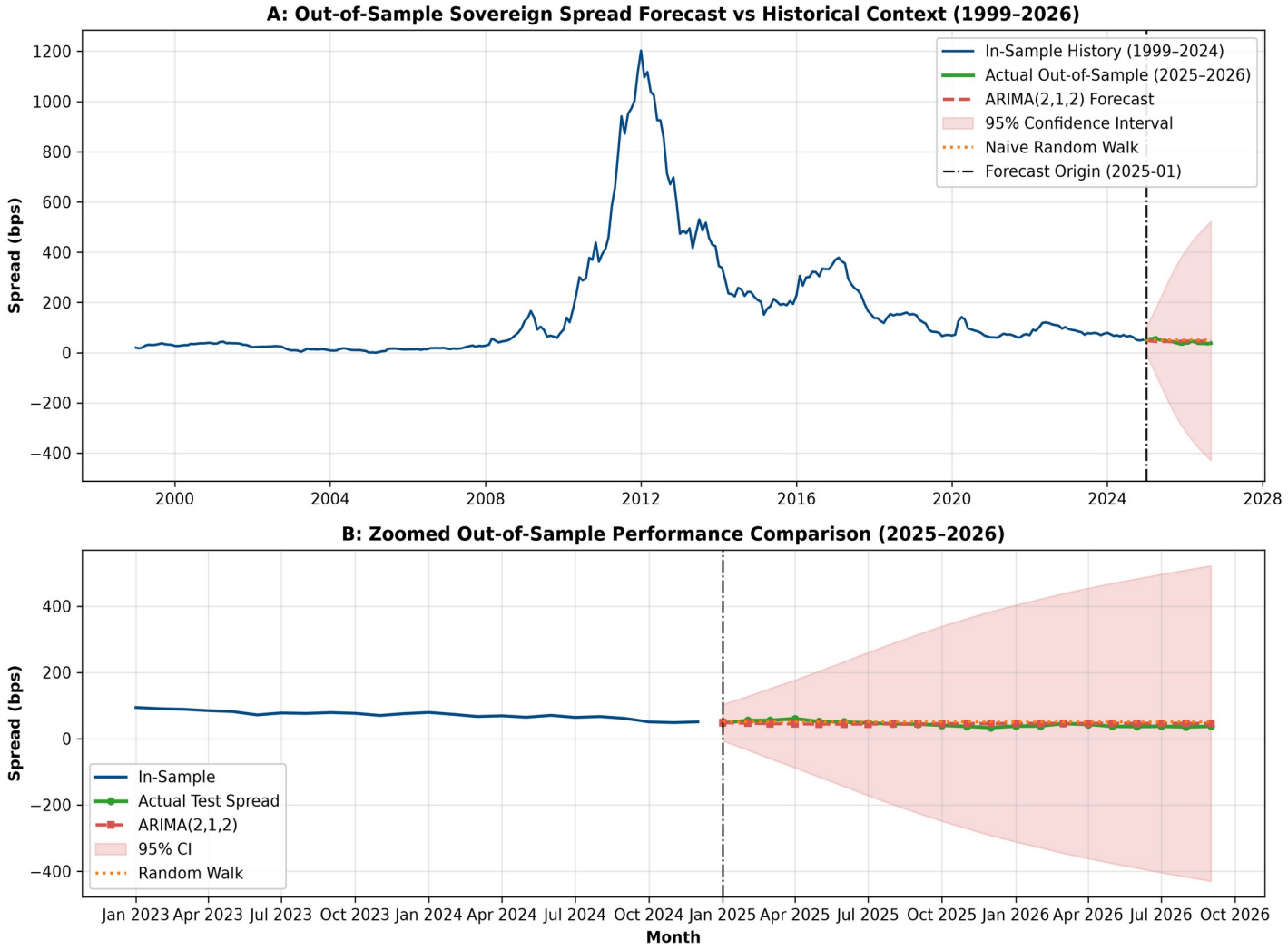
GARCH(1,1)-t Volatility Model

Parameter	Estimate	p-value
ω (Intercept)	0.794	0.145
α_1 (ARCH shock)	**0.174*** **	**0.0001**
β_1 (GARCH persist)	**0.826*** **	**< 0.0001**
ν (Student-t df)	**4.559*** **	**< 0.0001**
Persistence (α+β)	1.000	IGARCH
Engle ARCH LM	96.83***	< 0.0001

Cointegration & Volatility Insights:

- Cointegration Test: Engle-Granger p = 0.268 (Fail to reject H_0: No Cointegration). The long-run yield relationship broke in crisis; VECM is invalid, proving First Differences is required.
- Volatility Regime Shift: Conditional volatility peaked at 120 bps/month in 2011–12 and collapsed to under 12 bps under ECB backstops.

Out-of-Sample Forecasting: ARIMA(2,1,2) vs Random Walk (2025–2026)



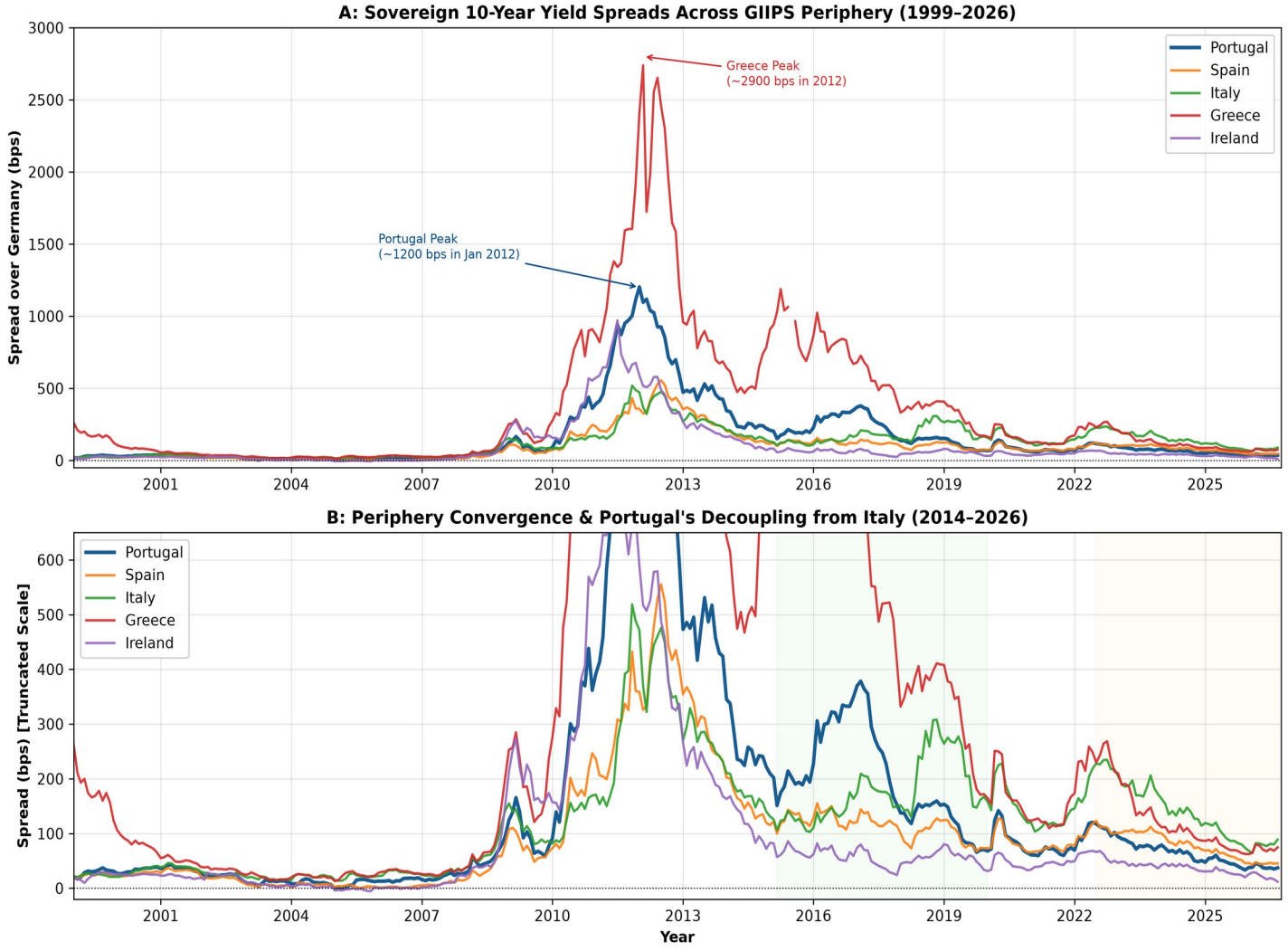
Out-of-Sample Accuracy (N = 21)

Model	MAE	RMSE	Theil's U
Naive Random Walk	8.91	10.22	1.000
RW with Drift	9.87	11.34	1.110
ARIMA(2,1,2) ★	**6.86**	**7.98**	**0.780**
ARIMA(0,1,1)	9.12	10.46	1.023

Key Forecasting Results:

- ARIMA Beats Random Walk: Theil's U = 0.780 confirms that ARIMA(2,1,2) outperforms the random walk by 22% in RMSE and 23% in MAE.
- Mean Absolute Percentage Error (MAPE) drops from 22.6% (RW) to 16.5%.
- Spread Mean Reversion: The model successfully captures the gentle upward mean reversion in 2025–2026.

Cross-Country Extension: Portugal's Decoupling from GIIPS



Cross-Country Divergence (bps)

Country	Crisis Peak	Pandemic	Latest (2026)
Portugal	**1,203 bp**	**71 bp**	**37 bp ★**
Spain	555 bp	81 bp	45 bp
Italy	519 bp	143 bp	89 bp
Greece	2,739 bp	137 bp	75 bp
Ireland	971 bp	41 bp	12 bp

The 'Portuguese Miracle':

- From Bailout to Tier-1 Periphery: In 2012, Portugal had the 2nd widest spread in Europe. By 2026, Portugal's spread (37 bps) is less than half of Italy's (89 bps) and below Spain (45 bps).
- Fiscal Prudence: Sustained primary budget surpluses and rapid debt reduction dismantled the peripheral penalty.

Conclusions, Policy Implications & Limitations

Core Conclusions

1. Regimes Matter More Than Levels: Sovereign bond pricing cannot be treated as a single linear process across 1999–2026. The 2012 structural break transformed market mechanics.

2. Backstops Alter Market Pricing: Prior to July 2012, public debt increases directly expanded spreads (+14.4 bps/pp). Post-2012, the ECB backstop neutralized default fear, shifting spread sensitivity to global risk appetite (VIX).

3. ARIMA Outperforms Random Walk: An identified ARIMA(2,1,2) specification achieves a Theil's U of 0.780, beating naive random walk forecasts.

4. Periphery Decoupling: Portugal transitioned from distressed Troika debtor to a resilient sovereign borrower trading inside Italy.

Policy Implications & Limitations

Policy: Credible Backstops are Crucial: ECB anti-fragmentation tools (OMT, PEPP, TPI) are vital to eliminate redenomination premia and prevent speculative self-fulfilling crises in currency unions.

Policy: Backstops Require Fiscal Virtue: ECB backstops provide liquidity insurance, but Portugal's decoupling demonstrates that domestic fiscal discipline (debt-to-GDP reduction) is essential for permanent spread compression.

Limitation: Data Frequency: Monthly frequency smooths high-frequency announcement effects; event-study methodology could capture daily market surprises.

Limitation: Interpolated Debt: Quarterly debt-to-GDP requires interpolation to monthly frequency, which may dampen short-term fiscal shock volatility.